

Computing cloud Fundamentals and Platforms

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Introduction

This Assignment will describe what are components needs to build a data centre on small premises and how to working some specification on cloud computing like redundancy, IaaS, PaaS and SaaS and as well as a little a bit about challenge face by IT department on cloud computing and restriction.

Components and prices

Blade servers:

Hewlett Packard
Enterprise

Hewlett Packard Enterprise ProLiant DL380 Gen10 server 72
TB 2.1 GHz 32 GB Rack (2U) Intel Xeon Silver 500 W DDR4-
SDRAM (P23465-B21)



General	
Processor family	Intel Xeon Silver
Processor frequency	2.1 GHz
Processor model	4208
Internal memory	32 GB
Internal memory type	DDR4-SDRAM
Memory layout (slots x size)	1 x 32 GB
HDD size	2.5 "
	Serial ATA, Serial
HDD interface	Attached SCSI (SAS)
Ethernet LAN	Y
Power supply	500 W
Redundant power supply	
(RPS) support	Y
Chassis type	Rack (2U)

Processor cores 8

Price: € 2,254.82 unit x 5 total €11,274.41

Source: <https://www.senetic.ie/product-p23465-b21/>

Sata Hard drive:



Western Digital.

Red Pro NAS Hard Drive WD2002FFSX 3.5" SATA 2,000 GB Hdd - 7,200 rpm 2 ms - Internal



Mfg. SKU: **WD2002FFSX**
SKU: 1041915 | EAN: 0718037835570

General
HDD size 3.5 "
HDD capacity 2000 GB
HDD speed 7200 RPM

[View full specification...](#)

€170.88

Excl. VAT: €138.93

- 1 + **ADD TO CART**

Availability: 31-40 pcs.

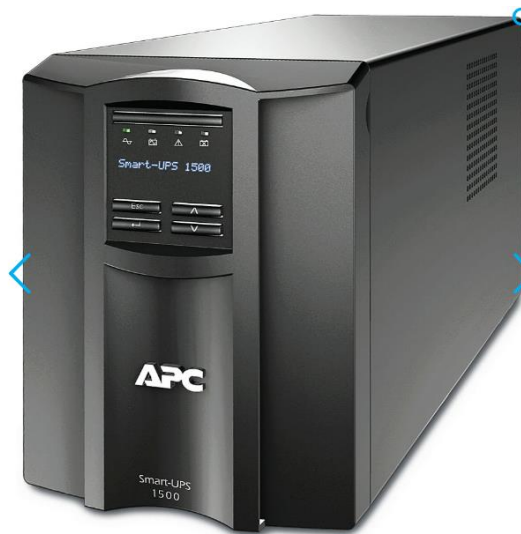
Features

HDD size	3.5 "
HDD capacity	2000 GB
HDD speed	7200 RPM
Interface	Serial ATA III
Type	HDD
Component for	NAS
Storage drive buffer size	64 MB
HDD interface transfer rate	6 Gbit/s
HDD sustained transfer rate	164 MiB/s
Start/stop cycles	600000
Mean time between failures (MTBF)	1000000 h

Price: € 170,88 unit x 4 total € 683.52

Source: <https://www.senetic.ie/product-p23465-b21/>

Power supply: UPS 1500 VA



€748.90

Excl. VAT: €608.86

-	1	+
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ADD TO CART

General

UPS topology

Output power

Output power capacity

AC outlet types

Power plug

AC outlets quantity

Battery technology

Battery life (max)

Battery recharge time

Form factor

Product colour

Cable length

Width

Depth

Height

Line-Interactive

1000 W

1.5 kVA

C13 coupler, IEC Jumpers

C14 coupler

8 AC outlet(s)

Sealed Lead Acid (VRLA)

5 year(s)

3 h

Mini Tower

Black

1.83 m

171 mm

439 mm

219 mm

Source: <https://www.senetic.ie/product-p23465-b21/>

Rack server:



units	Per unit
1 +	€515.40

Attribute	Value
Frame Type	4-Post
Rack Units	25U
Material	Steel
Colour	Black

This rack allows users to adjust mounting depth established on their requirements in demanding IT environments.

Source: <https://ie.rs-online.com/web/p/server-racks/>

Operating system:



Microsoft Windows Server 2016 Datacenter 16 Core Basic License



 Delivery time: 5-20 minutes
 Product key + download

€2,749.99 *

~~€3,000.00 *~~

Total price: 2,749.99 € *
Prices incl. VAT plus shipping costs

Item number: 96964622
Manufacturer No: P71-08653
Runtime: Lifetime
Language(s): All languages
Countryzone(s): Worldwide

Essentials Edition	Standard Edition	Datacenter Edition
Kleine Unternehmen mit Basisanforderungen an die IT; sehr kleine oder keine eigene IT Abteilung	Für alle Unternehmen die erweiterte Features benötigen und im geringeren Umfang virtualisieren	Für alle Unternehmen mit hohen Anforderungen an die IT Workloads mit großer Anzahl von virtuellen Systemen
25 User / 50 Devices - keine Server CALs erforderlich	Nicht limitiert, Server CALs erforderlich	Nicht limitiert, Server CALs erforderlich
1 physisch oder virtuell*	2 VMs	Unlimitierte Anzahl VMs
	2 Hyper-V Container**	Unlimited Hyper-V Container
	Unlimited Windows Server Container	
	Nano Server als Deployment Option	
Basis Funktionalitäten von Windows Server	Kernfunktionen von Windows Server	Storage Features inkl.: Storage Replica & Storage Spaces Direct
		New Networking Stack
Muss Root der Domäne sein		Shielded VMs und Host Guardian Service

VMs: virtual Machine

Source: <https://blitzhandel24.com/microsoft-windows-server-2016-datacenter>

The cost total of network equipment of a data centre on small premise is: € 15,972.22.

Step by step to build a data centre.

These steps that will describe below was thinking in how to developed an infrastructure of data centre on a small premises:

Power: First of all, when considering a build, a small data-centre on premise is essential to know if your environment has enough stable power to deal with a several server units in that case 5 blade servers. The IT department can be useful to determine the amount of energy that necessary to run the servers, but is always to ask a professional electrician for any maintenance.

Server backup: A back-up system it is needed to protect data in any case of an emergency, it could even be another data centre.

Cooling system: The point of the cooling system should be able to distribute the right temperature between the rack servers. Temperature should be regulated by IT department.

Installation of rack servers: The rack server must be installed leaving enough space between racks or cabinets inside the enclosure for air to circulate. Below the image How to set up a rack server:



Source: <https://www.serverroomenvironments.co.uk/blog/open-frame-it-cabinets-for-comms-rooms-and-datacentres>

Fire system: A fire system must be installed inside the enclosure as way to warn a fire. A fire department should determine which the better way to installed it.

What is the purpose of Load balancer?

Load balancer mean according to bender (2016):” To achieve high availability that’s sustainable as you grow.”

Load balancer whether provided by a stand-alone device or as a feature of an ADC, (Application delivery controller) facilitates this process by performing health checks on servers, detecting potential failures, and redirecting traffic as needed to ensure uninterrupted service.

What Consist a data centre?

Basically, a data centre as Amazon AWS, google cloud and Microsoft Azure cloud consist in networking room which include servers, switches and also cooling floor. According to cisco (2021)” Data centre design includes routers, switches, firewall, storages systems, servers, and application delivery controllers.”

Then if you go through inside of installation of data centre probably will finds some of theses equipment’s and storages systems that will describes in the next paraps:

Routers: is a device used in computer networks to forward information contained in data packets, providing connectivity between devices such as computers, smartphones and tablets, in LAN networks with the internet.

Switches: A switch is a device that simply connects all elements of your network. It acts as a bridge or control unit so that computers, printers, servers and all other types of devices can communicate.

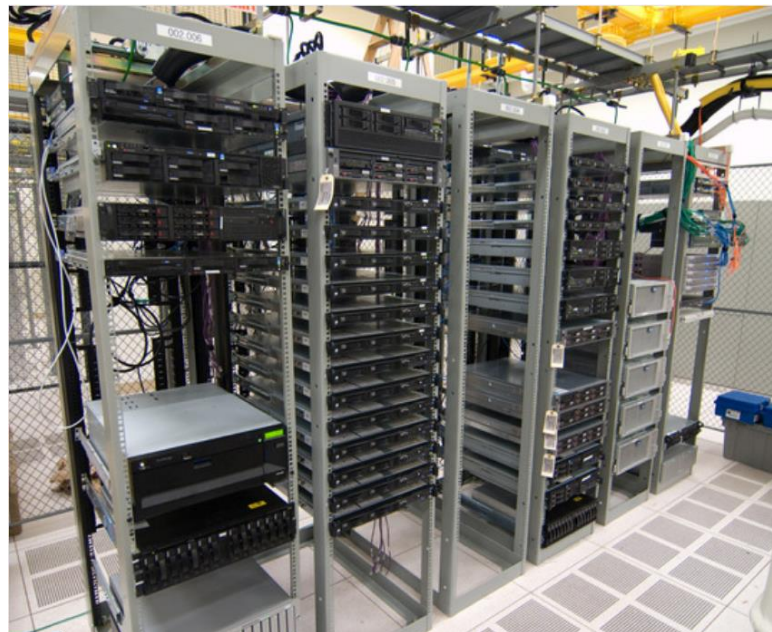
A switch works through A MAC (Media Access Control) address is the unique identifier that all hardware has assigned to it. They are trusted identifiers used to differentiate devices that send and receive data on the network.

Firewalls: Firewall system is a security solution based on hardware or software (most common) that, based on a set of rules or direction, analyses network traffic to determine which data transmission or reception operations can be performed. Its mission basically consists of blocking unwanted data traffic and freeing up welcome access.

Severs: A server is a computer equipped with one or more processors, memory storages, communication ports, software, and occasionally some data storage system, such as internal hard disks or SSD (Solid state drivers) memories.

Storage systems: large data centres market servers as services, renting virtual servers with processing, storage and communication resources to provide services through communication networks such as the internet. For example, Virtual machines, according to Hoffman (2020):" A virtual machine app creates a virtualized environment—called, simply enough, a virtual machine—that behaves like a separate computer system, complete with virtual hardware devices".

Infrastructure systems: A data centre infrastructure consists of all the physical components that make up and support a data centre. Servers are, of course, an important form of data centre infrastructure, but this category also includes things like network routers and switches, storage systems and firewalls, as well as server racks, redundant power supplies, and cooling devices. Below the images of server rack:



Source: <https://risingracks.com/blog/beginners-guide-to-server-racks/>

According to rack solution (2018)" A server rack is a structure that is designed specifically to house technical equipment including routers, switches, hubs, and of course, servers." In addition to securing the servers in place, the rack was set up also to provides an

easy flow of cool air from the systems that keeps servers and other network components at a constant temperature.

Cooling systems: Data centre cooling systems are designed to maintain temperature, humidity and air quality. Heslin says (2015):

Achieving this goal requires removing the heat produced by the ITE and transferring that heat to some heat sink. In most data centres, the operators expect the cooling system to operate continuously and reliably.

Controls the temperature of data centre is crucial because prevents of set of problems like,

- Excessive heat can burn equipment or cause machines to shut down.
- With this, servers can stop, causing the interruption of services that generates so much damage!

That is why it is essential that mission critical environments have a well-designed data centre cooling system.

What is Geo-redundancy and security restriction?

Data centre redundancy is essentially, the repetition of vital equipment that fulfil the same function. If one stops working, the other automatically takes over the task.

This is the case of power generators that replace a power grid in outages. Or redundant cooling units ready to take over the work in an unexpected failure. For example, data centre cooling systems.

The goals are to prevent any type of problem in equipment or in the energy supply from paralyzing the data centre, that would be called “Downtimes”. According to Moses (2020) “Data downtime refers to periods of time when your data is partial, erroneous, missing, or otherwise inaccurate, ranging from a few null values to completely outdated tables.

As much more layers of repetition, the greater the redundancy in the data centre and consequently, its reliability and security against failures and unscheduled outages.

To access to data centre some recommendation should be follow per person unauthorized and even some professional that work in a data centre:

- Security system fingerprint (professional to access some area restricted).
- To given a visitor badge in same case create a pin code.
- Surveillance Monitoring systems.

Hypervisor

A hypervisor, also known as a virtual machine monitor, is a process that creates and runs virtual machines (VMs). Tek- tools says (2020)“ A hypervisor is a process or a function to help admins isolate operating system and applications from the underlying hardware”.

in other words, virtualization is a software called hypervisor resides on the physical hardware to abstract machine resources which, in turn, are made available in virtual environments, which are called virtual machines.

The virtual machines on premises and in a cloud computing system works in the same way, the main difference is that virtual machines on cloud computing can be access of any computing and virtual machines on premises only could be access in a specific computer.

What are workloads?

Workload is the amount of processing the computer receives at any given time. The workload consists of a certain amount of application programming running on the computer and consistently a number of users logged and interacting with the applications on the computer.

A workload tool has the aim of programming triggers to trigger automatic workload processing, without the real need for a user interference. According to Bigelow:

Workloads are created to perform myriad different tasks in a countless variety of ways, so it is difficult to classify all workloads into a single set of uniform criteria. As one example, workloads may be classified as static or dynamic.

Increasingly computing claim are diverse, so than a lot types workloads were created those that come across in a hybrid cloud environment:

- **Batch Workload:** Describe the data that needs to be moved to a cloud for processing, how the data should be distributed, what parameters to use for each task, and

the command to start the process. According to McHugh: (2021)” Batch workload processing refers to groups of jobs[batches] that are scheduled to be processed at same time”. Currently batch workload still develops an important role in business, engineering, sciences and others areas that require the execution of the automatic tasks like, for examples processing invoices and payrolls.

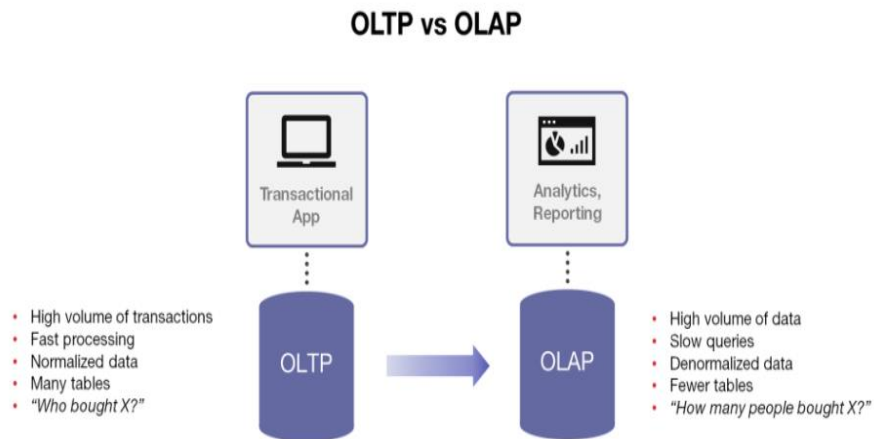
- **Transactional workloads x Analytic workloads:** It’s possible that some people confuse transactional and analytic workloads, according to (2020) Xie this happening due to:

In reality. it’s really based on the primary consumption pattern-are you building a system to facilitate “transactions” (think point-of-sale) or are you trying to “analyze” history to see trends and aggregated view?

The acronyms **OLTP** (Online transaction processing) and **OLAP** (Online analyse processing) are widely used in the Business intelligence (BI) universe. These acronyms help to understands how transactional and analytic workloads works. According to Allen (2021):

In the mid-1990s a clear split was acknowledged and discussed between classes of databases optimized for operational workloads, known as OLTP systems (online transaction processing), and databases optimized for analytical workloads, known as OLAP systems (Online analytical processing).

OLTP is the term used to refer to transactional systems, that is, the operating systems of organizations (company). They are used in the processing of routine data that are generated daily through the company’s information systems(on-premises) and support the organizational business execution functions. Next the image that show us the difference between OLTP and OLAP:



Source: <https://www.marklogic.com/blog/relational-databases-and-mixed-workloads/>

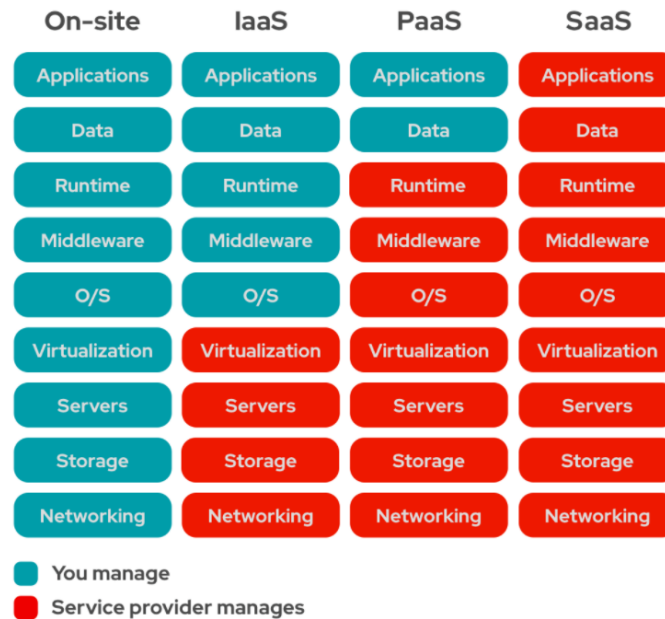
OLAP deals with the ability to analyze large volumes of information in the most diverse possibilities within a data warehouse.

- **High-performance computing workloads:** High-performance computing (HPC) is an essential tool to allow the acceleration of computational workloads, mainly in research. Cloud computing allow the flexibility need to work with HPC in the manner efficiently and cheaper if it is compared on-premise. According to: (2021)" High performance computing (HPC) is the ability to process data and perform complex calculations at high speeds".

- **Database workloads:** Techopedia says (2017) "Database Workload: Database can be analyzed for memory use, throughput at maximum loads and I/O throughput". There is situation that database workloads are time and self-contained, nonetheless in some situation, the data stored in database workloads are extensive, and the execution requires a specified and practical approach. For example, High-performance computing workloads database can be appliance on hardware's operating systems to give support the company or organization requirement.

IaaS, PaaS and SaaS

IaaS, PaaS and SaaS are nothing more than service models offered by cloud computing providers. In simple words companies that provide remote use of basic resources, these features come to replace the traditional infrastructure where the responsibility to keep the services running is now up to the provider(vendor) and no longer the customer. In the image below will be show how IaaS, PaaS and SaaS work:



Source: <https://www.redhat.com/en/topics/cloud-computing/iaas-vs-paas-vs-saas> 13/10/21

On – site (on premise): The infrastructure is provided by the own company such as physics (Hardware, room, tables computers etc...) and software (programs, networks and etc..).

IaaS (Infrastructure as a service): Basically, this service model allows you to migrate your company’s entire physical infrastructure to a large computing cloud, ensuring total autonomy and flexibility to change (increase or reduce) the resources hosted there. Servers, processing memory and storage can be changed according to your demand and need. According to Watts (,2019) “IaaS is fully self-service for accessing and monitoring computers, networking, storage, and other services.” The biggest example IaaS’s service currently is Microsoft Azure, Oracle Cloud, Amazon Web Services (AWS), IBM cloud and Google compute Engine.

PaaS (Platform as service): Is an intermediary model, where a more robust platform is provided to the customer while the infrastructure part is already prepared to receive software, whether its own or from third parties. According to Watts (2019) PaaS is:

Cloud platform services, also known as Platform as a Service (PaaS), provide cloud components to certain software while being used mainly for applications. PaaS delivers a framework for developers that they can build upon and use to create customized applications.

Solution’s examples of PaaS’ s services are Heroku, Windows Azure Cloud, Oracle Cloud, Amazon Web Services (AWS), IBM Bluemix and Google Cloud.

SaaS (Software as a service): As the acronym itself show us, it works as software as service. Among its functions are access to data via the internet, centralized management, integration and customization capabilities, among others. Watt (2019)” SaaS utilizes the internet to deliver applications, which are managed by a third-party vendor, to its users”.

SaaS is perhaps the cloud computing category best known to common users. Although not very explicit, there are several applications that already use this cloud computing model in the market. The most known example among people SaaS are: Dropbox, Google Drive, Salesforce and Netflix.

Understanding Paas vs IaaS and SaaS

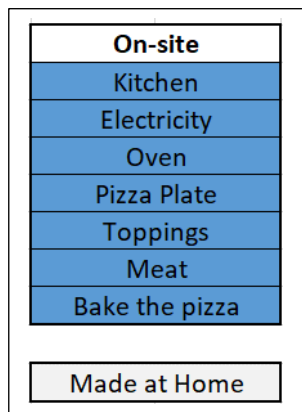
In the way to try to make simpler to understanding the example below it was creating based in the concept of Cloud computer in relation to PaaS, IaaS and SaaS compared with a restaurant the bake pizza it was thought to bake a pizza as- service. In the next images will be shown how it work:

Bake the pizza as a Service

On-site	IaaS	PaaS	SaaS
Kitchen	Kitchen	Kitchen	Kitchen
Electricity	Electricity	Electricity	Electricity
Oven	Oven	Oven	Oven
Pizza Plate	Pizza Plate	Pizza Plate	Pizza Plate
Toppings	Toppings	Toppings	Toppings
Meat	Meat	Meat	Meat
Bake the pizza	Bake the pizza	Bake the pizza	Bake the pizza
Made at Home	Kitchen-as-a Service	Restaurant	Pizza-as-a Service

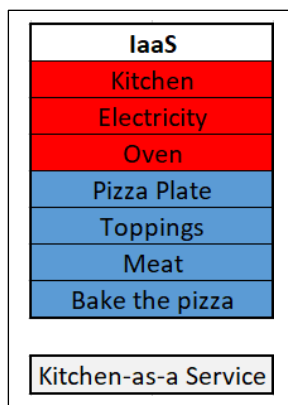
■ Managed by you
 ■ Managed by provider
 ■ Layout

Made at Home (On-site): For Hardcore Pizza makers, everything is done from scratch, choosing the ingredients to the mode of preparation and full control of the pizza toppings. below the image of On-site (infrastructure restaurant) work:



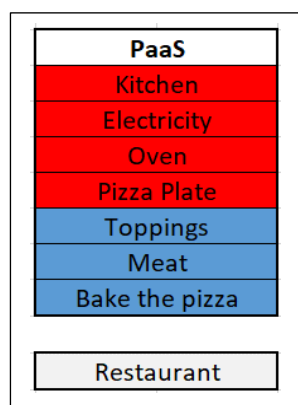
This is similar to traditional method where all the hardware and software is built by company's choice of requirements.

Kitchen as a service(Fully personalized,IaaS): Kitchen equipment is the infrastructure and is where the code runs, then the next step is to you to buy your own ingredients and make the lasagne as per the specification.



This is similar to the IaaS where the infrastructure is provided and the rest is managed by you.

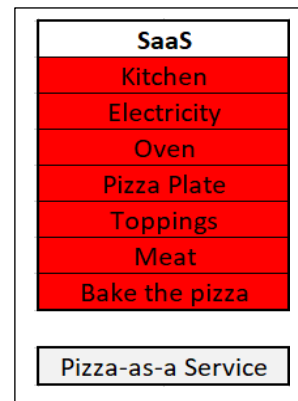
Walk to The Restaurant(Semi personalized,PaaS): It's ready to cook and where you walk into to the restaurant's kitchen to bake and it's not need to be concerned about buying ingredients, prepar work or even having the suitable tools.



This is like the PaaS where the infrastructure and hardware or software is provided, then the rest is up to the You .The Components are modified as resquest (change the image).

Pizza as a Service (no Personalized, SaaS): Here where you are able to enjoy the whole pizza (a fully finished product) once order. It's possible to call the restaurant where the pizza is already drafted, prepared and baked by someone else and will arrive in your home hot and ready to eat.

This is like the SaaS works where the deployment and framework the project is already set.



Migration to cloud computing

For companies considering their first move to a cloud (in this case DigiTech) that they would like to know some information about migrate some of their virtual machine and website, there are many factors that should consider from the benefits and risks to the model and types of on-premises cloud services for your business.

The migrating to the computing cloud platform may bring many solutions and benefits:

- The application is increasing facing an increase in traffic and are becoming dynamically resource-hungry to meet growing demand.
- It needs to reduce operating costs and increase the effectiveness of IT processes.
- It is becoming more difficult and expensive keep up with growing storage needs.

Potential risk must be considering to moving to the cloud platform, there are some general disadvantages associated with cloud migrations that should be considered:

- If the hardware is controlled by someone else, there can be some loss of transparency and control when debugging performance issues.

- Unwanted “neighbours” can hamper performance through shared resources that some cloud providers provide.
- Downtime. This happens to everyone, but you may not want to feel that your availability is controlled by someone else.

I would recommend IaaS (Infrastructure as a service) It is the best option for companies that need to maintain their own logical and application support. Consuming for example the infrastructure of a third-party data centre and working on top of virtual machines (VMS) but still able to outsource the service of your operating system to focus more on development, deployment and monitoring.

Challenge Task

Like every technological innovation that cloud computing has brought, in addition to the advantages, there are also concerns and challenges to be overcome by cloud engineers. For example, access to internet, old systems and security risk.

Access to internet: Cloud computing includes all computing services, such as database and storage, through the internet. So, it needs a good working of the network to have a good performance.

This is absolutely one of the main challenges faced by the IT sector. This is because, even with the expansion in internet access, there are still data packages that work with flaws and have a high cost. That's not counting the network instabilities.

Old systems: Many environments and systems need specific hardware or software to make them work. And, in many cases, these platforms were designed to respond particular needs. Not to mention that they were created at a time when cloud computing didn't exist yet. So, it's critical to assess whether the company's system can be accomplished in the cloud.

It may be necessary to make a diagnosis of what kind of changes must be made, if the platforms are accustomed for virtualized and if they fulfilled the basic requirements for the working of cloud computing.

Security risk: Information security must be a significant concern about hiring a cloud service provider. Mainly due to the complexity of data protection and

privacy regulations that many businesses held. To certify compliance with these aspects, technology development should be considered. For example: encryption, token validation and other solutions that assure the transfer of sensitive data stored in the cloud.

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